SOP-220

Compressor — Discharge Pressure High Response

Document ID	SOP-220
Revision	Rev 2.0
Document type	Operating procedure
Effective date	2026-03-09
Next review	2027-03-09
Document owner	Operations Engineering, EastRefinery
Applicable site	EastRefinery
Applicable unit	Unit 3
Asset class	compressor
Classification	Internal

1 Purpose and Scope

This procedure defines the operator response to a Discharge Pressure High alarm on centrifugal compressors at EastRefinery, and the investigation route for repeated occurrences.

3 Immediate Response

1. Acknowledge the alarm and confirm the reading against the redundant transmitter.
2. Verify anti-surge valve position against its commanded position.
3. Confirm the discharge cooler outlet temperature is within band.
4. Check downstream valve line-up for an unexpected closure or restriction.
5. If pressure continues to rise, reduce suction throttling in 5 percent steps.

> **NOTE** A discharge pressure alarm that clears immediately on anti-surge valve movement indicates a control-loop cause rather than a process restriction.

4 Repeated Occurrences

Discharge Pressure High alarms that recur without a corresponding process change are most commonly caused by anti-surge controller tuning drift, fouling of the discharge cooler, or a downstream restriction that develops progressively.

Observation	Indicated cause
Alarm clears on valve movement, recurs within hours	Controller tuning drift
Cooler outlet temperature trending upward over weeks	Discharge cooler fouling
Alarm severity increasing over successive events	Progressive downstream restriction
Alarm correlates with ambient temperature	Cooler capacity limit

Table 4-1 — Recurrence cause discrimination

Where discharge pressure alarms correlate with suction-side alarms on the same machine, treat the suction condition as primary and investigate it first.

5 Escalation

Escalate to Rotating Equipment Engineering where the alarm recurs more than eight times in 30 days, or where any event is accompanied by a surge detection.